🚀 AI YouTube Content Repurposer – Workflow Documentation

📌 Overview

The **AI YouTube Content Repurposer** is an automation workflow built using **n8n** that converts a single YouTube video into multiple types of content using AI.

It automates the entire pipeline from **video input → transcript extraction → AI processing → multi-platform content generation → email delivery**.

🎯 Objective

To help:

- Content creators
- Developers
- Marketers

Quickly generate:

- LinkedIn posts
- Instagram captions + reel scripts
- Twitter threads
- Blog articles

From a single YouTube video.

⚙️ Workflow Architecture

```
Webhook → Extract Video ID → Fetch Transcript → Clean Data → AI Processing →
Content Generators → Merge → Email भेज
```

🧩 Node-by-Node Breakdown

1. 📥 Webhook Node (Input Layer)

Purpose: Receives user input

Input Format:

```json
{
  "youtube_url": "https://youtube.com/watch?v=VIDEO_ID",
  "email": "user@email.com"
}
```

---

2. 🧠 Function Node (Extract Video ID)

Purpose: Extracts `videoId` from YouTube URL

Output:

```json
{
  "videoId": "xxxxxx"
}
```

---

3. 🎥 HTTP Node (Apify – Transcript API)

Service Used: Apify Actor → `youtube-video-transcript`

Purpose: Fetch video transcript

Output:

- Raw transcript text

---

4. 🧹 Clean Transcript Node

Purpose:

- Remove unwanted XML/formatting
- Convert into clean readable text

Output:

```json
{
  "cleanTranscript": "..."
}
```

---

5. 🤖 AI Processing Node (NVIDIA LLM)

Model: `meta/llama-3.1-70b-instruct`

Purpose: Generate structured insights:

Output:

```
{
  "topic": "",
  "summary": "",
  "key_points": [],
  "hooks": []
}
```

✨ Content Generation Layer

6. ✍️ LinkedIn Generator

Purpose: Creates professional LinkedIn post

---

7. 📸 Instagram Generator

Purpose:

- Caption
- Reel script

---

8. 🐦 Twitter Generator

Purpose: Creates thread-style content

---

9. 📝 Blog Generator

Purpose: Generates detailed blog article

🔀 Merge Node

Purpose: Combine all generated outputs into single payload

🧠 Final Code Node

Purpose: Structure final response:

```json
{
  "linkedin": "...",
  "instagram": "...",
  "twitter": "...",
  "blog": "..."
}
```

📧 Email Node (Output Layer)

Purpose: Send all generated content to user email

Email Format:

- LinkedIn
- Instagram
- Twitter
- Blog

🔑 Required API Keys

1. Apify

- Used for transcript extraction
- https://apify.com

---

2. NVIDIA API

- Used for AI content generation
- https://build.nvidia.com

---

3. SMTP / Gmail

- Used for email delivery

🚀 Execution Flow

1. User sends YouTube link
2. Workflow extracts video ID
3. Transcript fetched via API
4. Data cleaned and processed
5. AI generates structured insights
6. Content created for each platform

7. All outputs merged
8. Sent to user via email

🎯 Use Cases

- Content repurposing
- Social media automation
- AI content pipelines
- Developer learning projects

⚠️ Limitations

- Depends on transcript availability
- API rate limits
- AI output quality varies

💡 Future Improvements

- Frontend UI
- Auto posting to platforms
- Multi-language support
- Database integration

📌 Conclusion

This workflow demonstrates how AI + automation can transform a single input into a complete content ecosystem.

It serves as a **blueprint for building scalable AI agents using n8n**.

👨‍💻 Author

Built by **Chandu (Shinka-6C)** 🚀